VOICE INPUT SIMULATOR FOR KEYBOARD

Submitted By

MUHAMMAD EHTSHAM
(Regd. No. 2023-uok-03922)

HAMMAD SABIR
(Regd. No. 2023-uok-03914)

SANA ULLAH
(Regd. No. 2023-uok-03920)

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University of Kotli Azad Jammu and Kashmir
Faculty of Computing & Engineering
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A Report

Submitted in partial fulfillment of the requirements for the 5th semester project of

Operating system

In

Information Technology

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University of Kotli Azad Jammu and Kashmir
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KEYBOARD SOFTWARE WITH VOICE CAPTURE

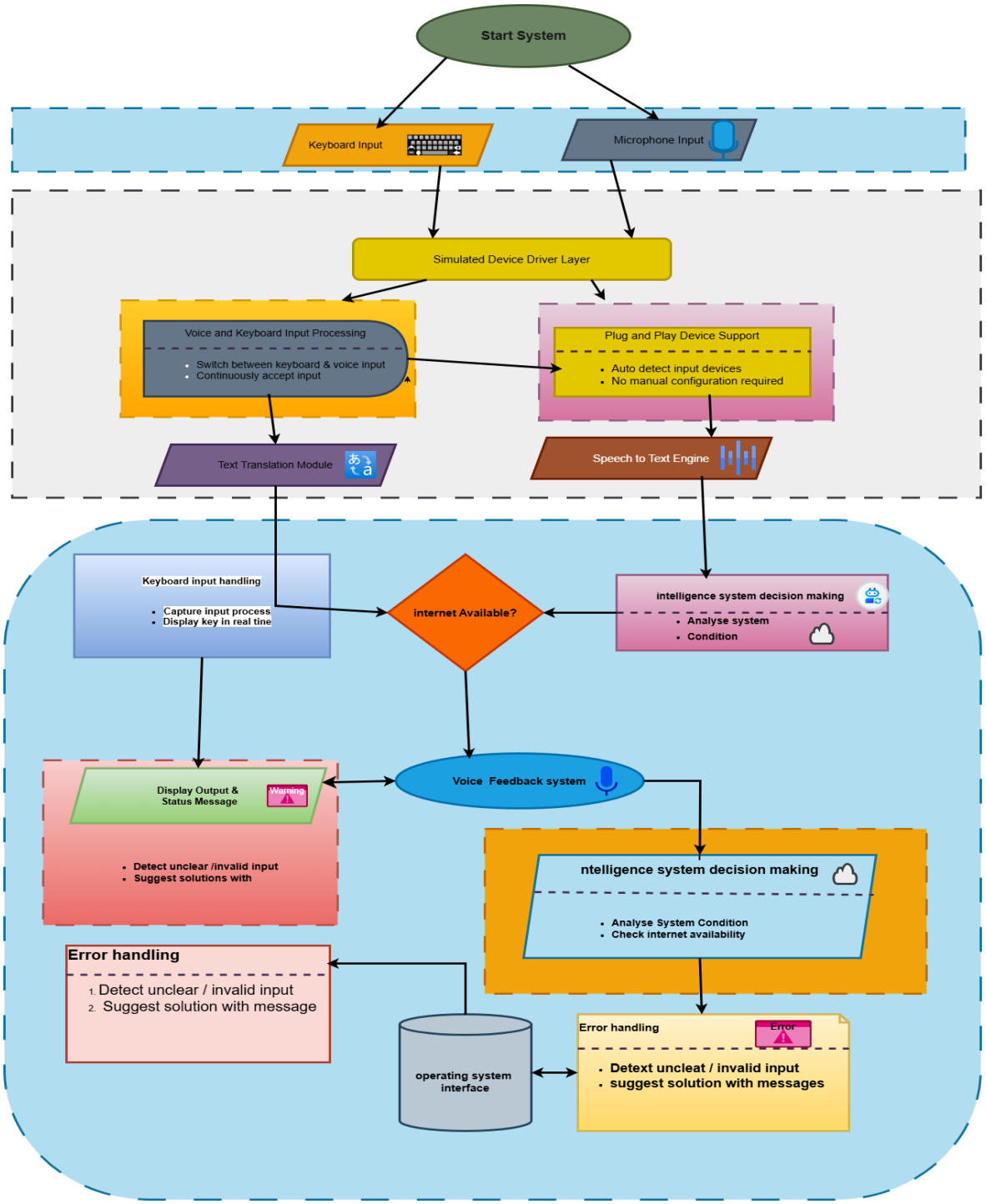

Figure 1.1 Keyboard Device Driver

1. Keyboard Software with Voice Capture

1.1 Introduction

This document explains the system diagram titled “Keyboard Device Driver Simulation with Voice Capture.”

The system demonstrates how a simulated device driver handles both keyboard and microphone inputs, processes them intelligently, and provides output to the operating system. It shows the interaction between hardware components, processing modules, decision-making systems, and the OS interface.

1.2 Objective

The main objectives of the system are:

- Simulate dual-input handling (Keyboard & Voice)
- Convert voice into text using a Speech-to-Text Engine
- Implement intelligent decision-making logic
- Provide real-time output and feedback
- Include strong error detection and handling
- Demonstrate interaction with the Operating System Interface

2. System working sequence

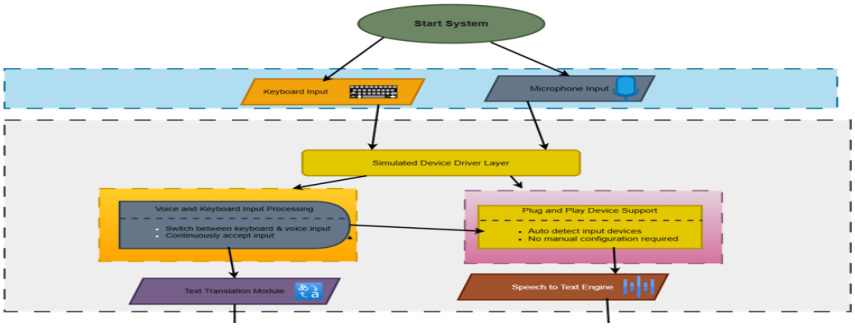

Figure 1.2 System Input

2.1 Start System

- Initialize system services and simulated drivers
- Activate keyboard monitoring
- Activate microphone monitoring
- Prepare communication with the OS

2.2 Plug and Play Device Detection

- Automatically detect connected keyboard and microphone
- Configure devices without manual setup
- Ensure driver readiness for input handling

2.3 Input Layer

2.3.1 Keyboard Input

- Capture typed keystrokes
- Convert them into digital key codes
- Send data to the simulated driver layer

2.3.2 Microphone Input

- Capture voice commands
- Convert sound waves into raw audio data
- Send audio stream to the simulated driver

2.4 Simulated Device Driver Layer

- Acts as a bridge between hardware and software
- Standardizes both keyboard and voice input
- Mimics real operating system driver behavior
- Forwards processed signals to the input processing module

2.5 Voice and Keyboard Input Processing

- Detects whether input is from keyboard or microphone
- Switches seamlessly between input sources
- Monitors continuous input
- Routes data to the appropriate processing path

2.6 Speech-to-Text Engine

- Converts voice/audio data into text
- Enables voice commands to behave like typed input
- Ensures high accuracy for reliable system execution

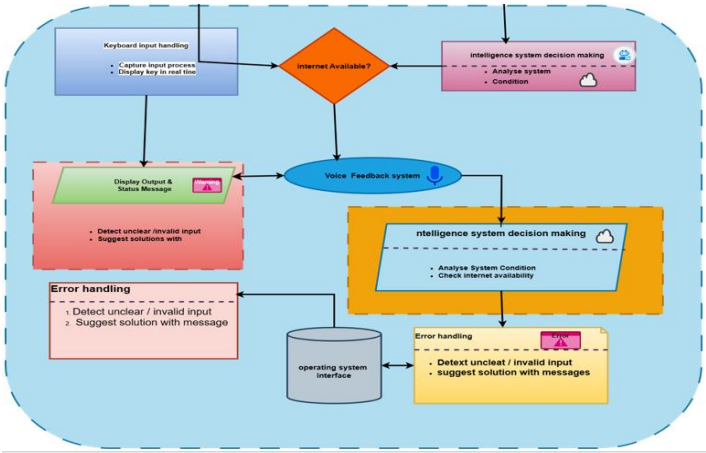

Figure 1.3 Processing And Output

2.7 Keyboard Input Handling

- Captures keystrokes in real time
- Displays typed characters
- Buffers input data
- Sends finalized text to the Intelligence Decision Module

2.8 Text Translation Module

- Translates processed text into other languages
- Enhances usability in multilingual environments
- Demonstrates system scalability and extensibility

2.9 Intelligence System Decision Making

- Analyzes received input
- Validates whether input is clear or invalid
- Determines required action
- Checks system conditions
- Proceeds to network availability verification

2.10 Internet Availability Check

- Decision module verifies network connectivity
- If Internet is available → enable online features (cloud translation, cloud speech processing)
- If Internet is unavailable → route to local processing
- Ensures uninterrupted system operation

2.11 Error Handling

- Detects unclear or invalid input
- Generates meaningful error messages
- Suggests corrective actions
- Improves system reliability and user experience

2.12 Voice Feedback System

- Provides audio confirmation
- Reports successful actions
- Announces detected errors
- Guides user through system interaction

2.13 Display Output & Status Message

- Displays final processed output
- Shows system status updates
- Indicates successful execution or errors
- Provides suggestions if needed

2.14 Operating System Interface

- Simulated driver communicates with the OS
- Sends processed input events to the operating system
- Integrates system responses into host environment
- Ensures smooth execution within OS architecture

2.15 Complete Workflow Summary

The system follows this structured sequence:

Start System
→ Plug & Play Detection
→ Input Layer (Keyboard / Microphone)
→ Simulated Device Driver
→ Input Processing
→ Speech-to-Text (if voice)
→ Keyboard Handling
→ Translation (optional)
→ Intelligence Decision Making
→ Internet Check
→ Error Handling
→ Voice Feedback
→ Display Output
→ OS Interface

2.16 Conclusion

This system presents a complete workflow of a Keyboard Device Driver Simulation enhanced with Voice Capture.

It demonstrates:

- Multi-input handling (Keyboard & Voice)
- Intelligent processing and decision-making
- Automatic device detection (Plug and Play)

- Robust error handling and feedback system
- Seamless communication with the Operating System

The structured sequence makes the system easy to understand and suitable for a 5th-semester project design documentation.